◆Rental Price Prediction Using Multiple Linear Regression

1. INTRODUCTION :-

The real estate and rental housing market has grown rapidly due to increasing urbanization and population. Determining the correct rental price of a property is often challenging because it depends on several factors such as area, number of bedrooms, bathrooms, location, parking availability, and property age. Incorrect pricing can lead to financial losses for property owners or make it difficult for tenants to find affordable housing.

Machine Learning provides an efficient way to estimate rental prices by analyzing historical property data. Among various machine learning algorithms, **Multiple Linear Regression (MLR)** is widely used because it predicts the rental price based on multiple independent variables. This project develops a rental price prediction system using Multiple Linear Regression to provide fast, accurate, and reliable rental price estimates.

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2. PROBLEM DEFINITION :-

Property owners and tenants often face difficulties in estimating a fair rental price. Traditional methods depend on market surveys, brokers, or personal experience, which may produce inconsistent and inaccurate results.

The main objective of this project is to develop a machine learning model using **Multiple Linear Regression** that predicts the rental price of a house based on features such as:

- Area (Square Feet)
- Number of Bedrooms
- Number of Bathrooms
- Parking Spaces
- Property Age

- Location (optional)

The system aims to reduce manual effort and provide accurate rental price predictions based on historical housing data.

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3. EXISTING SYSTEM :-

In the existing system, rental prices are usually estimated manually by property owners or real estate agents. These estimates are based on personal knowledge, nearby property comparisons, and market trends.

✓Limitations of the Existing System

- Manual estimation is time-consuming.
- Prices may vary depending on the broker or owner.
- Human errors can lead to incorrect pricing.
- Difficult to analyze multiple property features simultaneously.
- Lack of data-driven decision making.
- Inconsistent and less accurate rental price predictions.

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4. PROBLEM STATEMENT

The rental housing market is highly dynamic, and determining a fair rental price is a challenging task for both property owners and tenants. Rental prices are influenced by multiple factors such as property area, number of bedrooms, number of bathrooms, parking availability, property age, and location. Traditional rental price estimation methods rely on manual analysis, personal experience, or real estate agents, which can often result in inconsistent, inaccurate, and time-consuming decisions.

This project aims to develop a Rental Price Prediction System using Multiple Linear Regression that can accurately estimate the rental price of a house based on its key features. By analyzing historical housing data, the system learns the relationship between property characteristics and rental prices, enabling it to predict the expected rent for new properties. The proposed system provides a fast, reliable, and data-driven solution that helps property owners set competitive rental prices and assists tenants in making informed rental decisions.

5. PROPOSED SYSTEM :-

The proposed system uses **Multiple Linear Regression**, a supervised machine learning algorithm, to predict rental prices automatically.

The system collects historical property data, preprocesses the dataset, trains the regression model, and predicts the rental price based on user inputs. Users enter property details such as area, bedrooms, bathrooms, parking, and property age, and the trained model instantly estimates the expected rental price.

✓Advantages of the Proposed System

- Fast and automatic rental price prediction.
- Higher accuracy compared to manual estimation.
- Reduces human effort and errors.
- Uses multiple property features for better prediction.
- Easy to use through a simple interface.
- Helps both property owners and tenants make informed decisions.
- Can be improved further by adding more features like locality, amenities, and market trends.

✓Tools & Technologies Used

- Programming Language: Python

- Machine Learning Algorithm: Multiple Linear Regression
- Libraries: Pandas, NumPy, Scikit-learn
- User Interface: Tkinter or Flask
- IDE: VS Code, PyCharm, or Jupyter Notebook